Dataset	Train inputs	Test inputs	# Actions
Query-Keyword	1.23M	6.2K	33.2M
ORCAS	1M	8.5K	1.07M

Table 1: Statistics for the Query-Keyword datasets from a commercial search engine and the public ORCAS dataset.

consider an independent test set for evaluation. As in prior work [4], the candidate pool of items remains the same across train and test datasets, typically of the order of millions of items.

Datasets. On the query-document matching task, we use a dataset from a commercial search engine and a public dataset (see Table 1). We also use a Amazon dataset for product recommendation.

- **Query-keyword recommendation**. This dataset contains queries and ad keywords from a commercial search engine's logs. Given a query, the goal is to predict ad keywords that share the same intent or a more general intent than the query (i.e., *'phrase match'*). The base model is a 4-layer bi-encoder model that has been finetuned with contrastive loss on more than 20M query-keyword pairs. The training data consists of 1M new query-item pairs, along with a candidate pool of over 33 million items.
- **ORCAS** [3]. This public dataset contains clicked webpages from Bing for queries from the TREC deep learning challenge. For our evaluation, we consider the search query as the user input and the webpage title as the item. We filter the dataset to remove click data where either the query or the webpage title are empty. The dataset contains 17.5 million query-webpage pairs. To simulate a production setup, we utilize the majority (16.5M) of the dataset for training a supervised model, that acts as the base model. The base model is initialized with SimCSE [8] and trained for 5 epochs. The remaining 1M are used as *new* training data for optimizing novelty. We also reserve a separate, randomly sampled test set consisting of 8.5K query-keyword pairs.
- **AmazonReviews** [18]. This dataset contains users' product review histories on Amazon. Based on a user's previous reviews, the goal is to predict the next product that they would review. Li et al. [18] convert it to text-based problem by represent items as a text sequence using their metadata (e.g., "Title: Philips motor Category: Home Appliances Color: Black"). Users are represented as a text concatenation of each item in their profile. For our experiments, we consider the *Industrial & Scientific* domain consisting of over 11K user histories and 5K products. Li et al. [18] sort each user's history by time and break it down into a train set, validation set (second-last product), and a test set (most recent product in the history). We use the train set for training the base model, which is initialized with the pre-trained *RecFormer* model from [18]. To simulate the production scenario where users review additional items over time, we use augmented review history for finetuning novelty models that includes both the train and validation set products (*Finetuning* dataset). In both cases, the test set remains identical and we ensure that the test set is never used during training (since the finetuning stage does not have a validation set for early stopping; we train models for a fixed number of epochs).

Relevance Feedback. For all datasets, we use GPT-3.5 as the reward model for providing relevance feedback during training. For the first dataset where the goal is produce keywords with more general intent than the query, we use the prompt,

> *Given the query "{Query}", is the following query expressing a similar but more general intent, "{Keyword}"? Please answer in a single word: Yes/No.*

For the ORCAS dataset where the goal is to predict the top search result titles, we use the prompt,

> *Given the query, "{Query}", is it possible that the document titled, "{WebPageTitle}", is relevant for the user's intent. Please answer in a single word: Yes/No.*

The system prompt is the same in both cases, *"You are an expert in understanding user intent from search engine queries."*. For the AmazonReviews dataset, we use the system prompt, *"You are an expert in understanding user interests from Amazon.com product browsing data."*. The user prompt is given by,

> *A user is browsing Amazon.com for the following products:*
> *{ListOfProducts}*
>
> *Is this a relevant product that targets the same scientific interest?*
> *Recommendation: {CandidateProduct}*
>
> *Provide a brief reasoning and then the final answer within "⟨Yes/No⟩" tag.*

Metrics. We evaluate PG-Ret on the following offline metrics.

- **Novelty@k**: Novelty of top-k items compared to the base model, as defined in Definition 1. For the query-keyword dataset, we use L=200 for the base model since each retrieval model sends roughly 200 keywords to the downstream ranking layer. For ORCAS and AmazonReviews dataset, we use L=50 since we observe that the relevance of predicted items decreases significantly beyond 50 items.
- **Recall@k** is the number of clicked <query, item>pairs from test data that are in the top-k predictions. As novelty is optimized, recall over the top-k items should not decrease substantially.
- **Precision@k**: While recall is an important metric, it is dependent on the available clicks in the test data for each query. It is possible that a model makes relevant predictions but they are not counted since those <query, item>pairs do not exist in the test data. Hence, we use an advanced LLM, GPT-4 as the relevance evaluator for top-k items. Note that we use a different LLM for evaluation than the one used in training because 1) A more capable model like GPT-4 can provide more reliable relevance feedback; 2) using a different model ensures fairness of evaluation when compared to other baselines. Moreover, we use standardized prompts that are used in the production system for evaluating retrieval models. For each task, these prompts have been validated against human-labelled data and they achieve more than 85% accuracy, thus serving as a reliable approximation of human feedback. The first task uses a prompt tuned for estimating phrase-match relevance of keyword and the second

Model	Novelty@50	Novelty@100	Novelty@200	Recall@100	Recall@200	Precision@3
Base Model	–	–	–	45.0	54.1	63.7
Supervised Finetuning	6.6	22.7	79.1	**49.1**	53.5	65.5
PG-Ret (Conservative)	14.5	39.6	107.8	46.8	**56.4**	**70.0**
PG-Ret (Aggressive)	**32.8**	**70.2**	**151.8**	27.6	37.7	51.8

Table 2: Novelty, recall and precision for PG-Ret compared to supervised finetuning. PG-Ret (Conservative) obtains substantially high novelty with equal or better precision. PG-Ret (Aggressive) provides even higher novelty but with reduced accuracy.

Query	Top-5 Ad Keywords (Base Model)	Top-5 Ad Keywords (PG-Ret (Conservative))
tankless water heater electric home depot	home depot tankless water heater residential electric tankless water heater electrical tankless water heater electric tankless water heater water heater electric tankless	**instant water heater electric** **instant electric water heater** **domestic electrical appliances** **instant water heaters electric** **instant electric water heaters**
file management pc	computer file management file management in windows file management file management software file management system	computer file management files management software **software to organize files** file management software **files software**
hayan sunrise senior living	sunrise senior living home sunrise seniors living sunrise senior living wheaton il sunrise senior care living who owns sunrise senior living	**living facilities for seniors** **senior living facilities** **elder living** **elderly living** **elderly living facilities**
can i finance my child's college with a home equity loan	home equity loan years dcu home equity loan home equity financing 10 year home equity loan home equity loan in texas	home equality loans **educational financing** **household finance** **home equity laons** home equity laon

Table 3: Example predictions from base model and PG-Ret. Bolded items are novel compared to top-200 from the base model.

task uses a prompt tuned for estimating the general relevance of a query and a webpage title.

In addition, we evaluate PG-Ret using an A/B experiment for the first task of matching query to relevant ad keywords.

Baselines. For each dataset, we compare PG-Ret to a supervised model initialized with the base model and trained on the same training data using the InfoNCE loss with random negatives (Eq. 1). All models are trained using Adam optimizer with a learning rate of 10^{-5} and batch size of 128. PG-Ret (Aggressive) uses $M = 5$. For the AmazonReviews dataset, we use the same hyperparameters as in [18].

5.2 Query-keyword recommendation task

Novelty. For the query-keyword dataset, the goal is to increase the novelty of policy's recommendations. For training PG-Ret, we use the reward from Equation 7. The results are shown in Table 2. Compared to supervised finetuning on the same training set, PG-Ret leads to 2X-5X gains in novel keywords in top-50. In top-200, PG-Ret (Conservative) and PG-Ret (Aggressive) obtain 108 and 152 novel keywords respectively compared to 79 from the supervised finetuned model. At the same time, recall of PG-Ret (Conservative) is almost the same as the base model. In fact, recall@200 is slighly higher than the base model. To check the quality of the PG-Ret models, we also evaluate precision@3 as evaluated by GPT-4. While we would expect the precision to decrease due to the novelty loss, we find that PG-Ret (Conservative) has a substantially higher precision than the base model. This may be possible because the novelty loss can encourage the model to move away from local minima and sometimes find a more optimal solution. In comparison, PG-Ret (Aggressive) suffers a significant drop in both recall and precision, indicating that novelty optimization has led to a decrease in the model's accuracy. The offline results indicate that PG-Ret (Conservative) is a good balance between novelty and accuracy.

Table 3 shows qualitative results for a sample of the queries where PG-Ret led to novel keywords in the top-5 predictions. The base model tends to match keywords based on lexical overlap whereas PG-Ret is able to find rephrases with the same meaning.

A/B test. Finally, we evaluate PG-Ret (Conservative) on real user traffic using an A/B experiment over a 10 day period. As mentioned before, the retrieval system is engineered such that keywords from a new algorithm (e.g., PG-Ret) are selected for downstream ranking

Model	Novelty@10	Novelty@20	Novelty@50	Recall@50	Precision@1
Base Pretrained Model	–	–	–	**81.0**	**70.0**
Supervised Finetuning (3 epochs)	0.007	0.04	0.65	79.7	69.9
Supervised Finetuning (10 epochs)	0.07	0.33	2.82	78.7	68.4
PG-Ret (Conservative)	0.17	0.47	2.64	79.8	68.7
PG-Ret (Aggressive)	**0.33**	**0.85**	**4.03**	78.3	67.9

Table 4: Novelty, Recall and Precision on ORCAS. PG-Ret obtains substantial gains in novelty with minimal accuracy drop.

Model	Novelty@10	Novelty@20	Novelty@50	Recall@50
Base Pretrained Model	–	–	–	21.9
Supervised Finetuning	1.9	5.8	24.4	**22.6**
PG-Ret (Conservative)	**3.6**	**9.5**	**32.4**	22.4

Table 5: Novelty and Recall on the AmazonReviews dataset. Compared to supervised finetuning, PG-Ret obtains substantial gains in novelty with almost the same recall.

layer (which in turn, may lead to user impressions) only if they are novel compared to the existing algorithm. By including PG-Ret in the retrieval pipeline, we observe a 1% increase in query-ad matching density, the average number of relevant ads selected per query (as determined by the downstream ranker). We also observe a 0.14% increase in coverage, the fraction of queries for which relevant ads are shown to users. This indicates that the novelty optimization can help to match ads to the queries that other algos are not able to match relevant ads with. Finally, we also observe a 0.26% increase in click yield, the number of ad clicks per search query, indicating the new keywords recommended by the novelty optimization are well received by the real-world users. While the absolute number may look small, an increase of 0.26% can lead to a substantial impact when scaled to millions of users.

5.3 ORCAS: Query-webpage matching

To simulate the production setting, we train a supervised model on over 16M <query, webpage title>pairs as the *base* model. Our goal is to produce top-k webpages that are novel wrt. top-50 webpages predicted by the base model, using a training set of 1M pairs. We first use the InfoNCE loss to finetune the base model over the 1M training dataset (Supervised Finetuning). PG-Ret model is finetuned using our proposed method using the base model as the initialization.

Table 4 shows the results. We report results for two supervised finetuning models (epochs 3 and 10). In both models, the recall and precision decreases compared to base model, indicating that the 1M train set may lead to overfitting. PG-Ret (Conservative) also leads to a drop in recall but the corresponding novelty is significantly higher than that of a supervised model with similar recall. At comparable recall, PG-Ret (Conservative) obtains Novelty@50 of 2.64 compared to 0.65 for the supervised model (3 epochs). Overall, PG-Ret (Aggressive) obtains the highest novelty—on average, there are 4.03 webpages in top-50 predictions of PG-Ret (Conservative) that did not exist in top-50 predictions of the base model. The novelty is significantly higher than a supervised model (2.82, 10 epochs) with similar recall. Recall and precision of PG-Ret models decrease slightly (up to 3%) compared to base model, which can be a reasonable tradeoff given the gains in novelty.

5.4 Amazon: User-Product recommendation

Table 5 shows the novelty and recall metrics for the AmazonReviews dataset. Both Supervised Finetuning and PG-Ret models are initialized with the base pretrained RecFormer model and trained on the finetuning set. Since these models are trained on an additional recent product from the user's history, both models obtain slighly higher recall than the base model. However, at the same recall, novelty of the PG-Ret (Conservative) model is significantly higher than the supervised model. On average, PG-Ret recommends 3.6 products in its top-10 list that are novel wrt. the top-50 recommendations from the base model, compared to 1.9 novel products for the supervised model. As in the previous datasets, PG-Ret leads to improved novelty while incurring a minimal loss in recall compared to the supervised model.

6 CONCLUSION

We presented a technique to optimize a non-differentiable, task-specific loss in information retrieval applications. We justified the binary-action formulation of the problem through theoretical and empirical results. On empirical recommendation datasets, the proposed technique leads to substantial gains in novelty of top-k items.

While we used the simple REINFORCE-based RL algorithm, future work can consider Actor-critic or proximal policy optimization algorithms for optimizing novelty and how they can be extended to large action spaces. Exploring robust reward functions in the presence of noisy LLM feedback is also an important future direction.

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